ML Hackathon: The Predictive Modeling Optimization Challenge

1. Introduction and Background

The modern chemical industry relies heavily on complex computational fluid dynamics and boundary value problem (BVP) simulations to design and operate chemical reactors. However, these rigorous physics-based simulations are computationally expensive and too slow for real-time plant optimization.

At modern continuous flow facilities, process engineers oversee highly sensitive reaction networks. They constantly face a bottleneck: predicting reactor behavior quickly when operating setpoints change, without having to wait for heavy differential equations to solve.

Your task is to bridge the gap between chemical engineering and computer science by building a Predictive Model (a data-driven machine learning surrogate) that can instantly predict a reactor's performance based on historical plant data.

2. The Chemical Process

The dataset is derived from a non-isothermal, continuous flow reactor operating in a real-world industrial setting. The reactor operates with the following series-parallel reaction network:

1. 1. Desired Reaction: A ----> B (rate constant: k1)
2. 2. Side Reaction: B ----> C (rate constant: k2)

The Challenge: The reactor involves competing reactions that are highly sensitive to operating conditions. The kinetic and thermodynamic behaviors of this system create complex, non-linear trade-offs. Your predictive model must independently learn these hidden relationships—balancing flow rates, concentrations, and thermal profiles—to accurately predict the final yield of the desired product (B).

3. The Objective

You are provided with historical data from the plant. Your objective is to train a machine learning model that can accurately predict the Overall Yield of Product B given a new set of operating conditions.

- Phase 1 (Model Training): Use the provided train_dataset.csv to build and validate your model locally.

- Phase 2 (Prediction): Process the unseen operating conditions in test_dataset.csv and generate your predicted yield for each of the test cases.

4. Dataset Description

You will be provided with two files:
- train_dataset.csv (150 rows): Contains the input features and the target variable.
Link: https://drive.google.com/file/d/15lmh-VS6jx0UgE11Rq_xsoggopuo3RQa/view?usp=sharing

- test_dataset.csv (50 rows): Contains only the input features.
Link: https://drive.google.com/file/d/1J1jFgaMVUT6CWz_FWEMboU9w7R9Hdono/view?usp=sharing

Features (Input Variables)

• flow_rate_L_min: Volumetric flow rate of the reactant mixture (L/min).
• concentration_mol_L: Inlet concentration of Reactant A (CA0 in mol/L).
• inlet_temperature_K: Temperature of the feed entering the reactor (K).
• length_m: The specific length of the reactor (m).
• jacket_temperature_K: The temperature of the external heating jacket (K).

Target (Output Variable)

• overall_yield: The final yield percentage of Product B at the reactor exit.

5. Submission Guidelines

Final submissions must include the predictions in the file format specified below via the Unstop platform by the end hackathon period.

- The file must be a single .csv named [TeamName].csv containing exactly 50 rows (matching the order of test_dataset.csv).
- It must contain exactly one column with the header overall_yield.
- Predictions must be continuous numerical values (floats) rounded to at least 3 decimal places.
- Crucial Rule: Only one final submission is allowed per team. You must thoroughly evaluate and cross-validate your model locally before making your final upload.
- Finalists Requirement: Shortlisted teams will also be required to submit their fully documented Jupyter Notebook (.ipynb) files containing their complete workflow prior to the final presentation.

6. Evaluation Criteria

This competition evaluates both predictive accuracy and engineering intuition. Brute-forcing mathematical algorithms without understanding the underlying chemical system will not win this hackathon.

Phase 1: Quantitative Shortlisting (RMSE)

All 50 test rows will be evaluated blindly against a hidden master solution containing the true physics outputs. The primary evaluation metric is Root Mean Squared Error (RMSE).

$$\text{RMSE} = \sqrt{ [(1/n) * \Sigma (y_i - \hat{y}_i)^2] }$$

Based strictly on this score, the Top 6 teams will be shortlisted for the final phase.

Phase 2: Final Pitch & System Understanding

The Top 5 shortlisted teams will be invited to campus to deliver their final pitch to the judging panel. You must be prepared to defend your model and demonstrate a deep understanding of the reactor system, proving your results are grounded in physical reality.

Judges will evaluate the innovation, robustness, and scalability of your model, along with your overall pitch, based on:

- Process Insight: Your ability to explain the physical and thermodynamic trade-offs happening inside the reactor.
- Innovation & Feature Engineering: How creatively and accurately you translated chemical engineering principles into mathematical features for your model.
- Robustness & Scalability: How your model handles the realities of raw industrial data, your strategies to prevent overfitting on a strictly limited dataset, and how effectively the model could scale to real-time plant operations.

The final hackathon winners will be decided by the judges based on a holistic combination of your Phase 1 predictive accuracy and your Phase 2 pitch.